

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: U – 238 (Nat. U, Dep. U)			FORMS: Soluble		
PHYSICAL CHARACTERISTICS					
HALF-LIFE: 4.47 billion years					
DECAY EMISSIONS					
The emissions below include Th-234 and Pa-234m daughter emissions					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV, Ave)	%	E (keV)	%
63	5	822	98	4198	78
13	4	52	70	4149	22
16	3	27	19		
92	3	71*	11		
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY					
C – level (low hazard): ≤ 20 mCi (20 kg)					
B – level (moderate hazard): > 20 mCi (20 kg), ≤ 1 Ci (1000 kg)					
A – level (high hazard): > 1 Ci (1000 kg)					
EXTERNAL RADIATION HAZARDS			INTERNAL RADIATION HAZARDS		
The figures below include Th-234 and Pa-234m emissions			Annual Limit on Intake: 10 µCi (Ingestion)		
Gamma dose rate, point source at 1 ft, 1 mCi:			0.8 µCi (Inhalation)		
0.17 mrem/h			The values above indicate the activity taken into the body that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE).		
Beta dose rate to skin, point source at 1 ft, 1 mCi:			In general, uranium compounds are considered more chemically toxic as a heavy metal than as radiotoxic. Care should be taken to prevent ingestion through good hygiene and use of fume hoods or glove boxes.		
400 mrem/h					
Contamination skin dose, uniform deposit of 1 µCi (U238 and daughters) per cm²:					
10000 mrem/h					
SHIELDING			DOSIMETRY AND BIOASSAY REQS		
The figures below include Th-234 and Pa-234m emissions			Urine assays may be required after large spills or contaminations.		
Gammas/X-rays:					
3 cm of lead will reduce the gamma dose rate by 90%.					
Betas/electrons:					
9 mm of plastic will absorb all emissions.					
Bremsstrahlung may be created and require additional shielding.					
SPECIAL PROBLEMS AND PRECAUTIONS:					
1. Recommended survey probe: PGM					
2. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often.					
3. Survey work areas before, during, and after work. Instrument and smear surveys are required.					
4. Contact Health Physics regarding waste disposal.					
5. Solidify liquid wastes in cement kits.					

References:

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- Johnson, T.E., Birkly, B.K. (2012). Health Physics and Radiological Health (4th ed.). Baltimore, MD: Lippincott Williams & Wilkins.
- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
- Peplow, D. (2020) Specific Gamma-Ray Dose Constants with Current Emission Data. *Health Physics*, 118(4):402-416; 2020.
- Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
- 10.CFR.20 – Standards for Protection Against Radiation (2019). Retrieved from <https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/>